

“Enhanced Mobility Device with Proximity Sensors Technology.”

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Abstract

Navigating environments poses substantial challenges for visually impaired individuals due to the rapid changes in their surroundings, making memorization difficult and increasing the risk of serious injury. This dependency on assistance from others highlights the limitations of traditional aids like walking canes, which require physical contact with obstacles and have a limited detection range of usually 1 metre. Moreover, using walking canes demands significant mental mapping of the surroundings and constant attention to tactile feedback. This paper presents the design and simulation of an Enhanced Mobility Device using Proximity Sensors, aimed at improving navigation for visually impaired individuals. The proposed device is a stick integrated with HC-SR04 proximity sensors and NE555 timers, which detect objects and alert the user through DC-operated motors that provide vibrations. This system is designed to enable users to better navigate their surroundings without the need for constant physical contact or intensive mental mapping. The circuit was designed, simulated, and tested in Proteus simulation software, yielding positive results. The Enhanced Mobility Device demonstrates significant potential to reduce dependency on others, increase safety, and provide a more efficient means of navigation for visually impaired individuals.

Keywords: Object Detection; Proximity Sensor (HC-SR04); NE555 Timer; DC Motor.

I. Introduction

Blind individuals often face significant challenges navigating their surroundings without secondary human assistance, particularly in unfamiliar environments where obstacles are not easily detectable. Traditional walking aids such as canes and walking stands have significant limitations. To address this critical need for enhanced mobility and safety, various technological advancements have been developed, including Smart Glasses for the Visually Impaired [1]. These glasses use recognition technology and convert it into auditory feedback, thereby assisting visually impaired users in interpreting their surroundings, which is achieved through image processing techniques. Another revolution is the LiDAR-Based Obstacle Detection and Distance Estimation in Navigation Assistance for the Visually Impaired [2], of which EfficientDet-LiteV4 is used for obstacle detection, and a depth map from the LiDAR is used to estimate object distance to different obstacles.

Augmented Reality Navigation Systems For The Blind [3] convert physical environments to digital information. This system combines cameras and sensors including software algorithms to create a detailed overview of the users' surroundings to provide navigational assistance through auditory feedback. Advancement of haptic aids, such as the Haptic Navigation Aid for Spatial Perception, led to the creation of The OptiBand, a wearable haptic navigation device designed to enhance spatial awareness for blind and visually impaired individuals. Equipped with LiDAR sensors, it detects objects up to 15 metres away and communicates their presence and location through haptic feedback [4]. The Smart Shoes Safety System for the Blind based on Internet of Things (IoT) Technology incorporates three ultrasonic sensors and a microprocessor to detect obstacles and measure distances, alerting the user through auditory signals when an obstacle is near. The system also includes water and flame sensors for detecting potential hazards and uses GPS technology [5].

Developing a proximity sensor stick for the blind incorporates these advancements by providing real-time feedback about obstacles in the user's path, significantly improving their quality of life. The focus of this project is to design and develop a lightweight, portable, and user-friendly proximity sensor stick capable of detecting obstacles up to 1 m, with a battery-rechargeable feature providing non-intrusive feedback for individuals with varying degrees of visual impairment. Consideration has been given to factors such as cost-effectiveness, durability, and ease of maintenance to ensure the device's practical and widespread use, enabling blind individuals to navigate confidently and independently in both indoor and outdoor settings.

II. Materials and Methods

This section provides an overview of the Enhanced Mobility Device's subsystems. Utilising proximity sensors (HC-SR04), the device is designed to scan and detect objects within the user's path. The system comprises six subsystems: power supply, sensor system, timing system, switching system, and feedback or output system, as illustrated in Figure 1.

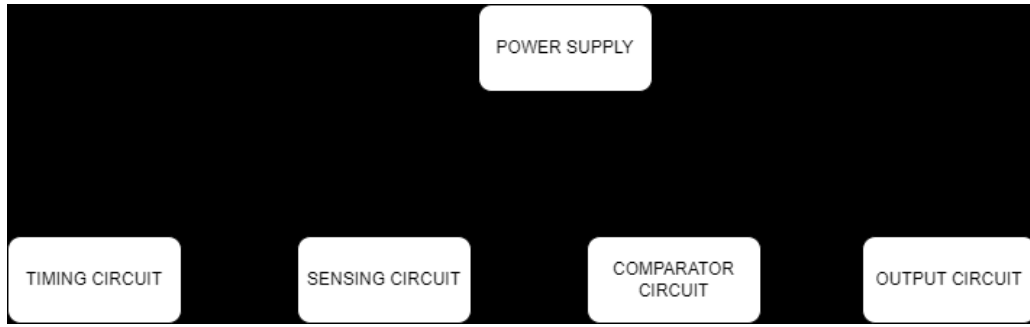


Figure 1: Block diagram for the Enhanced Mobility Device using Proximity Sensors subsystems.

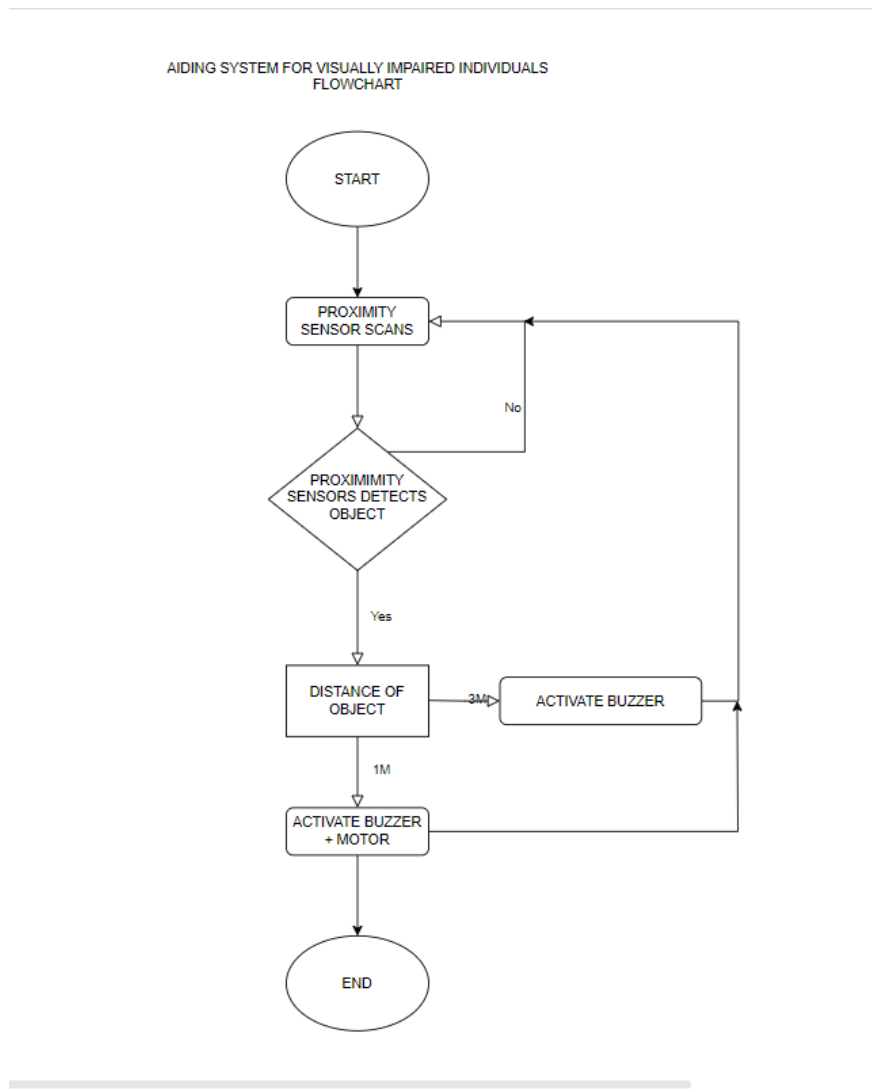


Figure 2: Flowchart for obstacle detection using the proposed design.

I. Power Supply

The design is operated by a 9 V battery. It supplies the different sub-circuits: timing circuit, comparator circuit, and the output circuit.

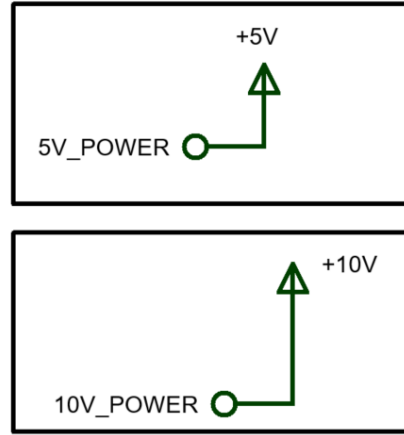


Figure 3: Diagram showing the power supply of the design.

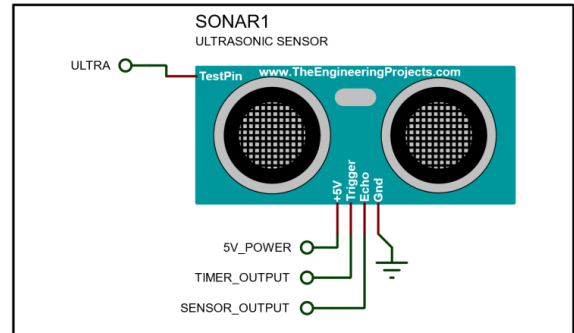
II. Sensory Circuit

The sensing circuit detects obstacles as the user is moving. For the design, the HC-SR04 ultrasonic component is used. The HC-SR04 sensor functions by emitting an ultrasonic pulse and receiving the echo of this pulse to measure distance. The primary components of an equivalent circuit include the trigger mechanism, the pulse generator (transmitter), the receiver, and the control logic that processes the echo time to determine distance. The ultrasonic distance measuring sensor is a non-contact distance measurement device. The range of measuring distance is 2 cm to 400 cm, with a precision of 3 mm. The internal structure of the HC-SR04 ultrasonic sensor mainly consists of three parts: the waveform generator, the signal receiver, and the control circuit. The control circuit primarily includes amplification, filtering, and shaping. The HC-SR04 sensor has four external I/O ports; the pin names are VCC, GND, Trig, and Echo. When the Echo pin receives the transmitted signal, the change of the pin becomes a high level, representing the success of a range measurement [6].



(a) HC-SR04 proximity sensor.

SENSORY CIRCUIT



(b) Proteus image of the sensory circuit.

Figure 4: HC-SR04 sensor and its Proteus simulation circuit.

III. Timing Circuit

For the design, input signals from the sensory circuits are sampled at fixed intervals. Instead of processing a single input, the system continuously scans and samples the surroundings for obstacles. The proximity sensor (HC-SR04), operating at 250 Hz, is configured using the NE555 timer in its astable mode to achieve the necessary functionality.

NE555 Timer in Astable Mode

The NE555 Timer IC configured in Astable Mode functions as a free-running oscillator, meaning it continuously generates a square wave output without requiring any external trigger. This mode is used to create clock pulses and PWM (Pulse Width Modulation) signals [7].

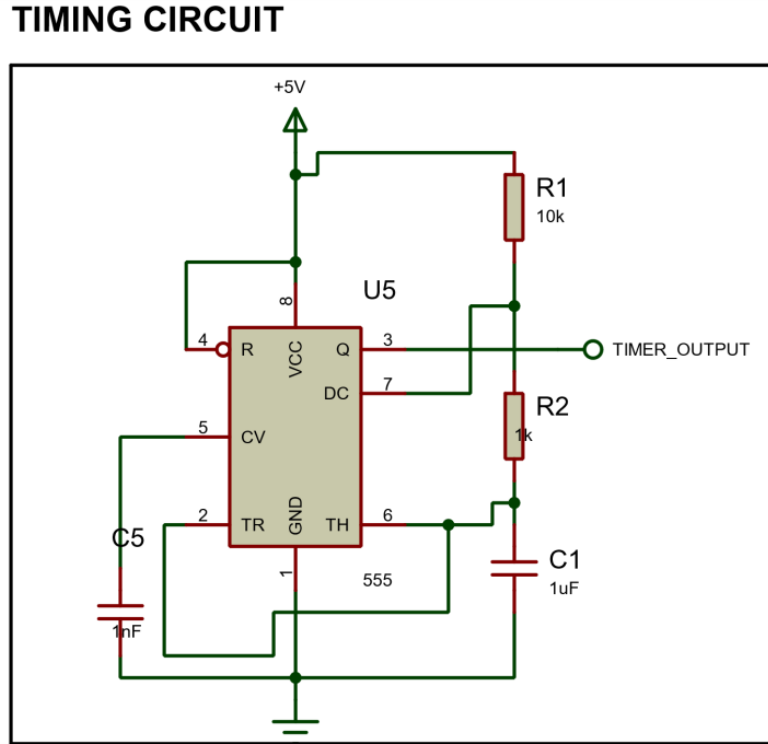


Figure 5: Internal diagram for the NE555 timer timing circuit.

To achieve a time period of 2 seconds, the following calculations are completed to determine the values of the capacitor and resistors R_1 and R_2 :

$$f = \frac{1.44}{(R_1 + 2R_2) \times C} \quad (1)$$

$$T = \frac{1}{f} \quad (2)$$

$$T = (R_1 + 2R_2) \times C \times 1.44 \quad (3)$$

where f is the frequency of oscillation, C is the capacitance in farads (F), and R_1 , R_2 are resistance values in ohms (Ω).

i. Frequency Setup:

Given the desired frequency $f = 250$ Hz, the combined product of resistance and capacitance is calculated as:

$$(R_1 + 2R_2) \times C = \frac{1.44}{f} = \frac{1.44}{250 \text{ Hz}} = 0.576 \text{ s} \quad (4)$$

ii. Capacitor Selection:

$$C = 1 \mu\text{F} \quad (\text{fixed value})$$

iii. Resistor Selection:

Choosing $R_A = 1,000 \Omega$ to balance the circuit design, R_B is calculated as:

$$1,000 + 2R_B = 36,000 \implies 2R_B = 35,000 \implies R_B = 17,500 \Omega$$

iv. Frequency Verification:

$$f = \frac{1.44}{(1,000 + 2 \times 17,500) \times 0.000001} \approx 250 \text{ Hz} \quad (5)$$

Therefore: $R_1 = 10 \text{ k}\Omega$, $R_2 = 1 \text{ k}\Omega$, $C = 1 \mu\text{F}$.

TIMING CIRCUIT

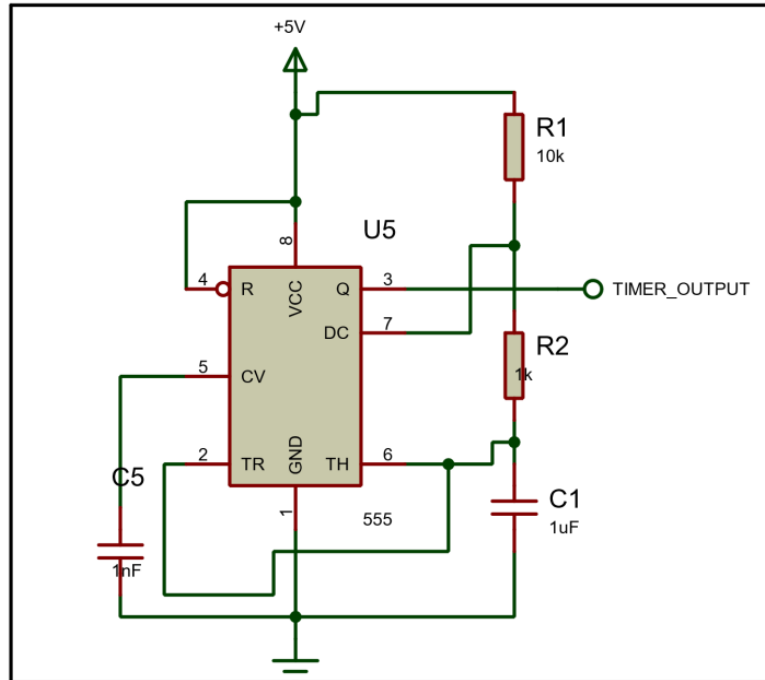


Figure 6: Timing circuit diagram (NE555 astable configuration).

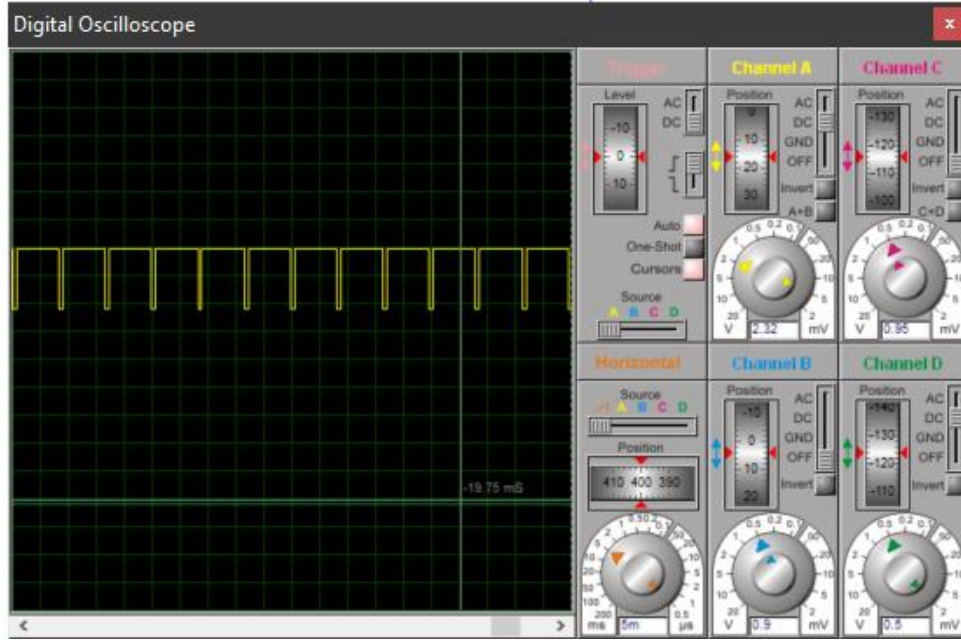


Figure 7: Oscilloscope image of the output signal from the NE555 timer.

IV. Digital-to-Analogue Signal Converting Circuit

The ultrasonic sensor (HC-SR04) outputs a digital signal, while the LM358 operates on analogue voltages. To interface these components, the digital signal must be converted to an analogue format using an RC low-pass filter [8, 9].

To design the filter, the frequency and voltage levels of the PWM signal from the ultrasonic sensor must be determined. The appropriate values for R_5 and C_6 are calculated to achieve a cutoff frequency of 1 Hz:

$$f_c = \frac{1}{2\pi RC} \implies 1 = \frac{1}{2\pi \times 1 \mu\text{F} \times R} \implies R = 200 \text{ k}\Omega \quad (6)$$

DAC CIRCUIT

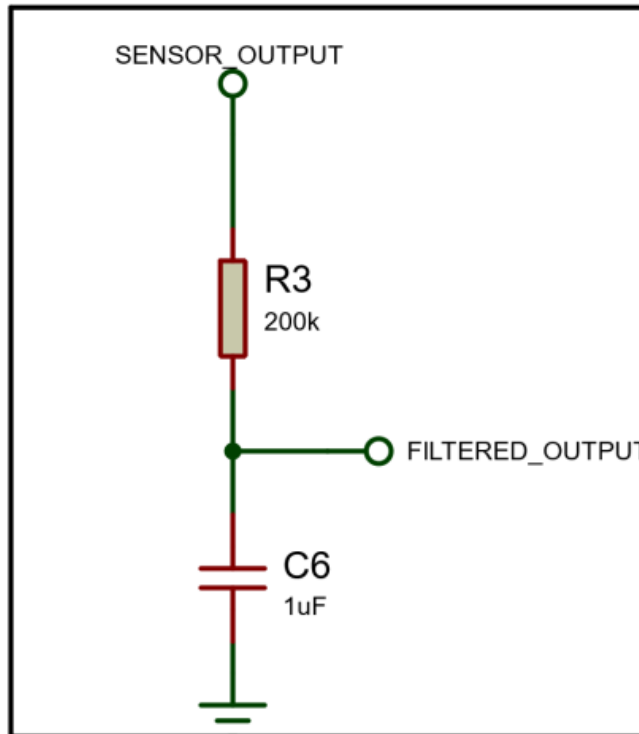


Figure 8: DAC (RC low-pass filter) circuit diagram.

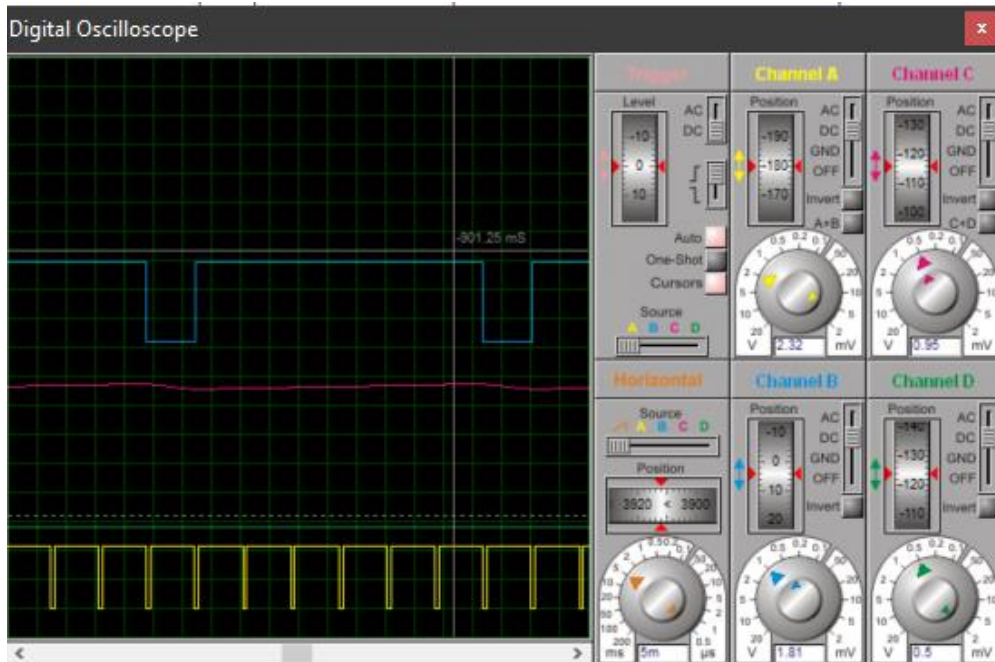


Figure 9: Oscilloscope image before and after filtering of the output signal from the ultrasonic sensor. The **blue** PWM signal represents the digital output from the HC-SR04 sensor (5 V / 0 V). The **red** signal represents the converted DC voltage analogue signal.

As evident, the output is initially in digital form (binary, switching between 5 V and 0 V).

The RC low-pass filter converts this signal to a DC analogue voltage proportional to the measured distance.

V. Comparator Circuitry

In this design, an LM358 operational amplifier is configured as a comparator to detect objects within 3-metre and 1-metre ranges using the HC-SR04 proximity sensor. The operational amplifier determines whether an object is present within the specified distances [10,11].

1. **Reference Voltage Setup:** The reference voltage on the inverting input of the LM358 op-amp is set slightly below the voltage corresponding to the target distance detected by the sensor. For example, if the sensor outputs 2.5 V at 3 metres, the reference voltage is set to approximately 2.4 V.
2. **Sensing and Comparison:** The voltage from the proximity sensor, which varies with the distance of the detected object, is fed into the non-inverting input (positive input) of the LM358. As the sensor detects objects, it produces a voltage proportional to the distance.
3. **Output Activation:** When the voltage at the non-inverting input exceeds the reference voltage at the inverting input, the output of the LM358 goes high (+VCC = 5 V), indicating that an object is closer than the threshold distance.

Comparator for 3-Metre Detection

The reference voltage divider is designed as follows:

$$V_{\text{ref}} = \frac{R_2}{R_2 + R_1} \implies 3.0 = \frac{R_2}{R_2 + 10 \text{ k}\Omega} \implies R_2 = 15 \text{ k}\Omega \quad (7)$$

Reference voltage: $V_{\text{ref}} = 3.0 \text{ V}$.

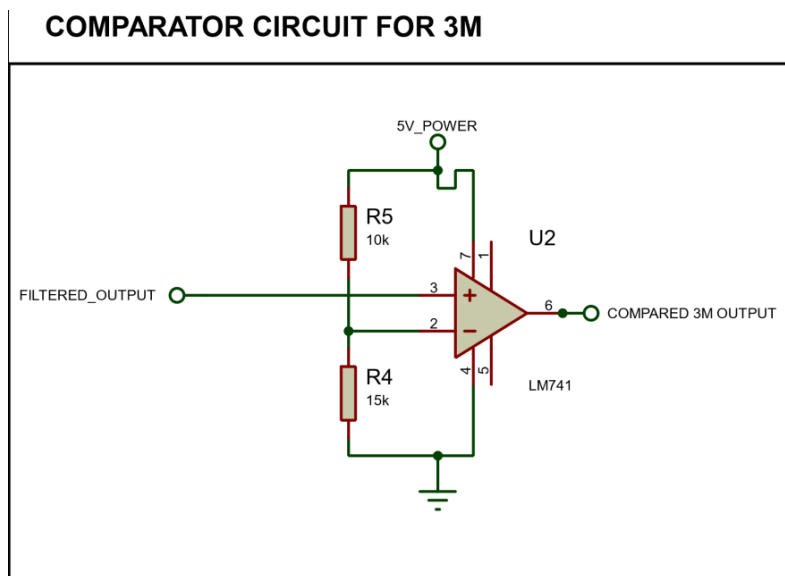


Figure 10: LM358 comparator circuit diagram for 3-metre distance alert.

Comparator for 1-Metre Detection

$$V_{\text{ref}} = \frac{R_2}{R_2 + R_1} \Rightarrow 1.5 = \frac{R_2}{R_2 + 10 \text{ k}\Omega} \Rightarrow R_2 = 4.3 \text{ k}\Omega \quad (8)$$

Reference voltage: $V_{\text{ref}} = 1.5 \text{ V}$.

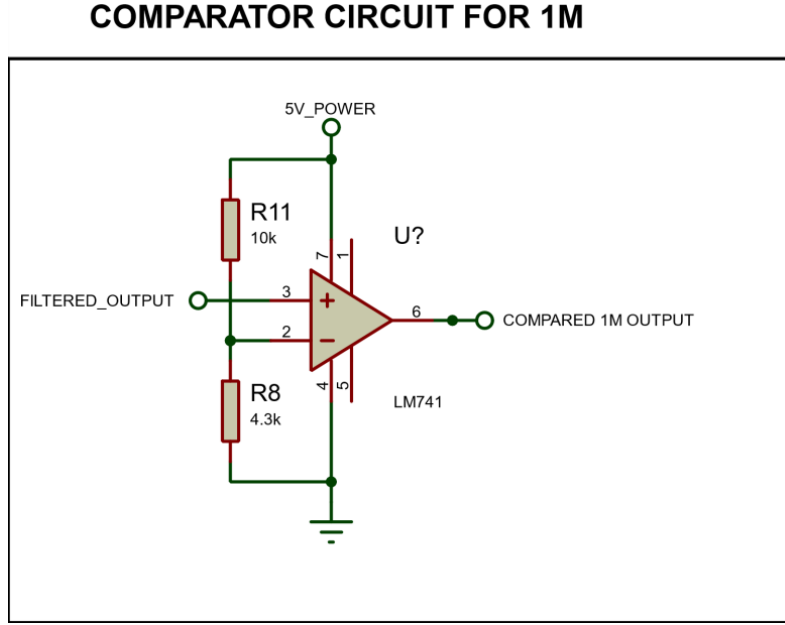


Figure 11: Comparator circuit diagram for 1-metre distance alert.

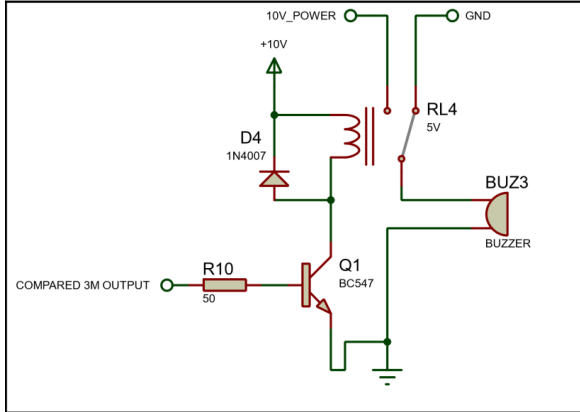
VI. Output Circuit

A relay switch (NT72C) is used to activate the output devices: a DC-operated motor and a piezoelectric buzzer. When current flows through the relay coil, it generates a magnetic field that attracts the armature and consequently switches the relay contacts. This mechanism allows the relay to open or close another circuit based on its normally open (NO) or normally closed (NC) configuration [9]. In this design, the normally closed position is connected to ground and the normally open position is connected to a 5 V power supply.

The output circuit feedback devices are:

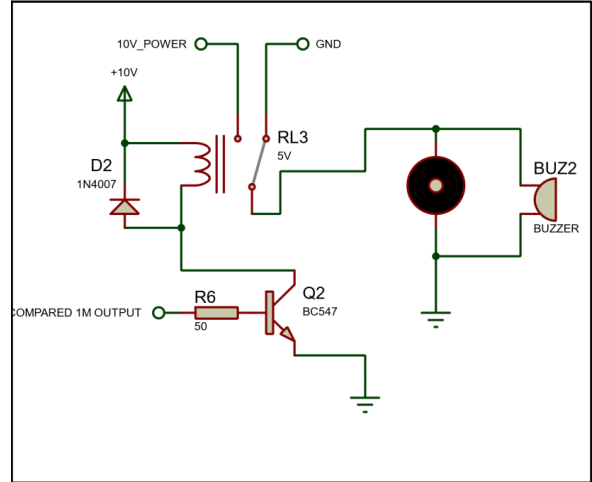
- A 5 V DC-operated motor and a piezoelectric buzzer, activated when the 1-metre LM358 comparator detects an object.
- A DC-operated piezoelectric buzzer only, activated when the 3-metre LM358 comparator detects an object.

OUTPUT SYSTEM FOR 3M



(a) Output circuitry for the 3-metre detection system.

OUTPUT SYSTEM FOR 1M



(b) Output circuitry for the 1-metre detection system.

Figure 12: Output circuit diagrams for both detection thresholds.

III. Results and Analysis

The device is designed to alert users as they approach obstacles from a specific distance. Initially, when an obstacle is detected 3 metres away, the device emits a sound from a buzzer to alert the user. As the user moves closer to the obstacle, reaching a distance of 1 metre, the device provides an additional alert through both a vibration from a DC motor and a sound from the buzzer. This dual-feedback system ensures that users are adequately informed about their proximity to nearby objects, enhancing their ability to navigate their environment safely.

Prototype

A working simulating prototype was developed in TinkerCad software and subsequently simulated on Proteus.

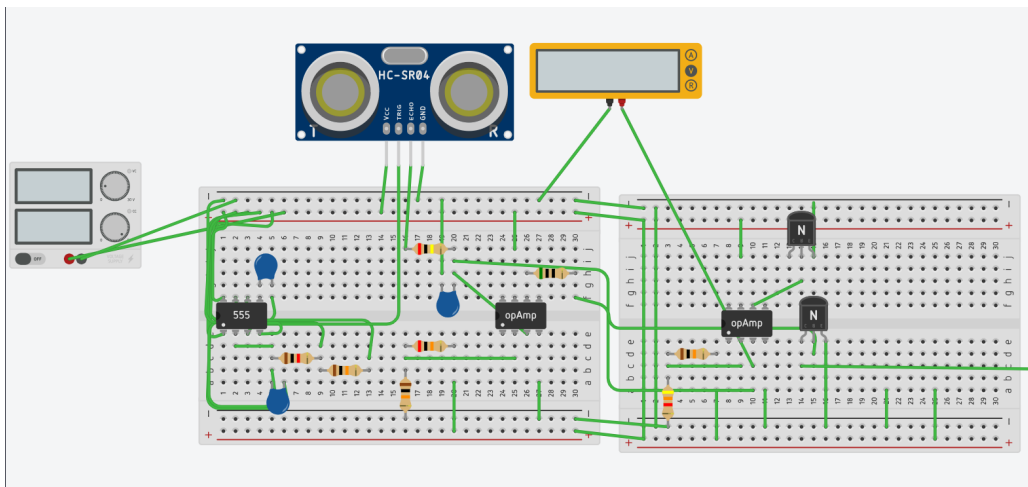


Figure 13: TinkerCad prototype showing the sensory, timing, and comparator circuits.

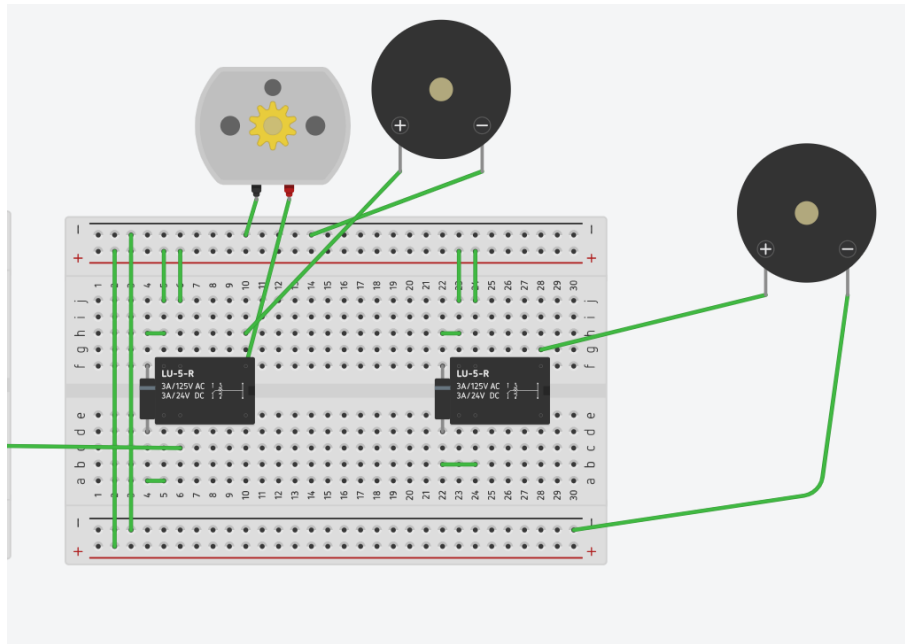


Figure 14: TinkerCad prototype showing the output circuit.

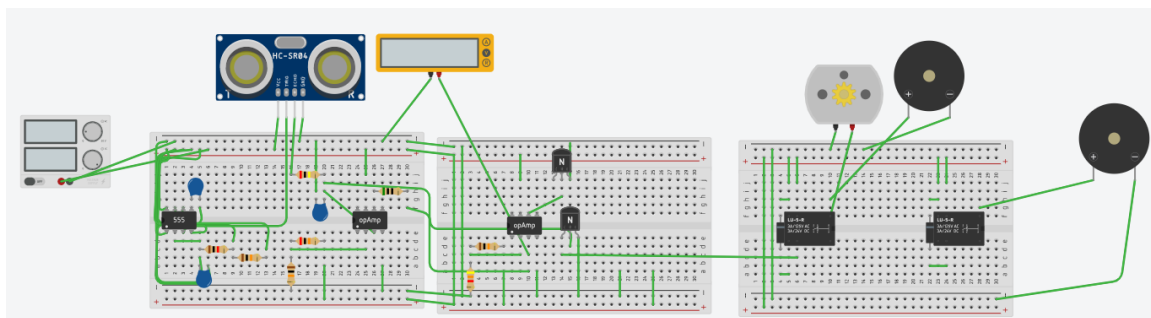


Figure 15: Overview of the complete TinkerCad prototype of the design.

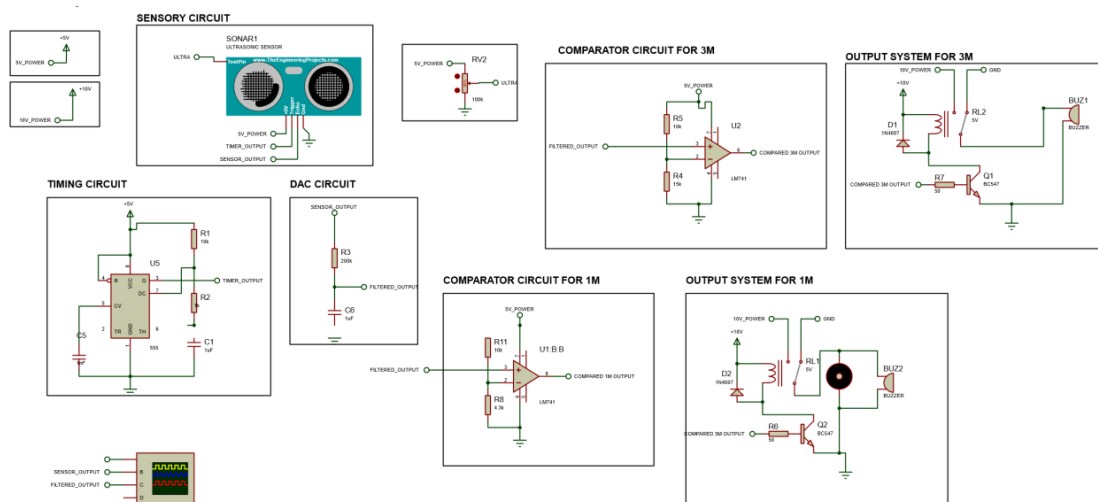


Figure 16: Proteus simulation of the complete circuit.

Testing

Off-State

When the system is powered up, the NE555 timer receives power and begins generating a square wave at 250 Hz. The HC-SR04 starts generating pulse waves of 8 oscillations. When there is no signal input to the sensor, no voltage drops are generated and the RC low-pass filter outputs 0 V at the node. This 0 V is conducted into both comparators (LM358). Since the reference voltage is higher than the voltage at the non-inverting terminal, the output voltage from both op-amps is 0 V. Consequently, the current at the base of the NPN transistors (BC547) is zero (cutoff state), no current flows at the collector, and the relay switch remains in the normally closed position. No current flows into the buzzer or motor, and no output feedback is produced.

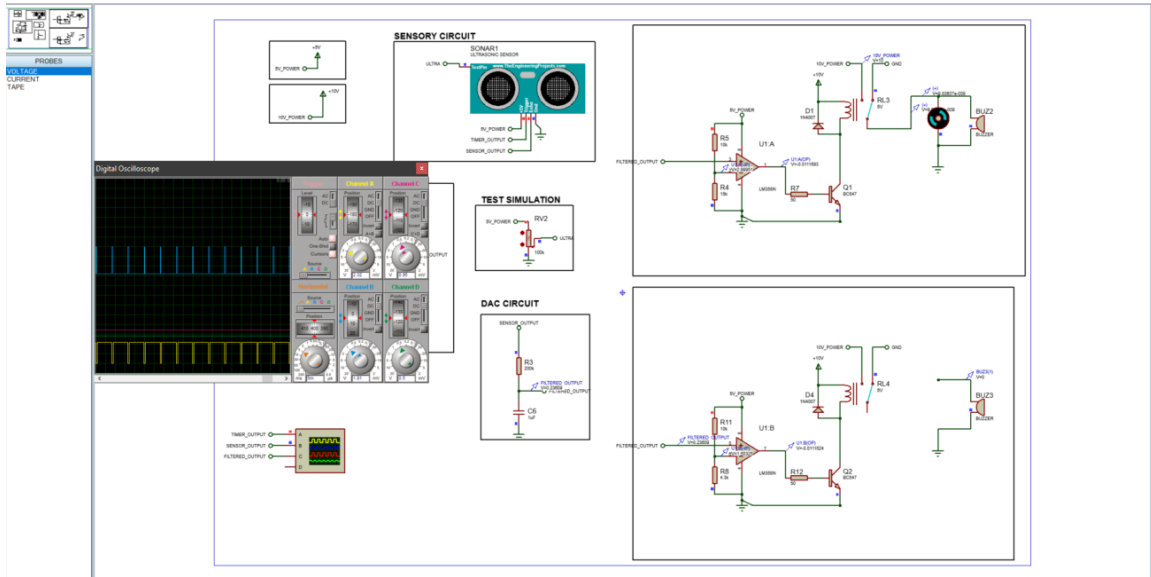


Figure 17: Diagram showing the different voltages at different areas of the circuit in the Off-State.

On-State

When the sensor detects an object and receives a trigger input, it generates a voltage in digital form. This voltage is converted to an analogue DC voltage via the RC low-pass filter and is conducted into the non-inverting terminals of the op-amps (LM358). When the voltage at the non-inverting terminal exceeds the reference voltage at the inverting terminal, the output of the LM358 goes to 5 V (VCC). This 5 V forward-biases the NPN transistor, allowing current to flow through the relay coil. The relay energises and switches from normally closed to normally open, activating the output devices. The specific output depends on the measured distance:

- **3-metre detection:** The buzzer produces sound.
- **1-metre detection:** Both the buzzer and DC motor are activated (sound and vibration).

The oscilloscope readings confirm that larger distances produce larger pulse widths, corresponding to higher analogue voltage levels.

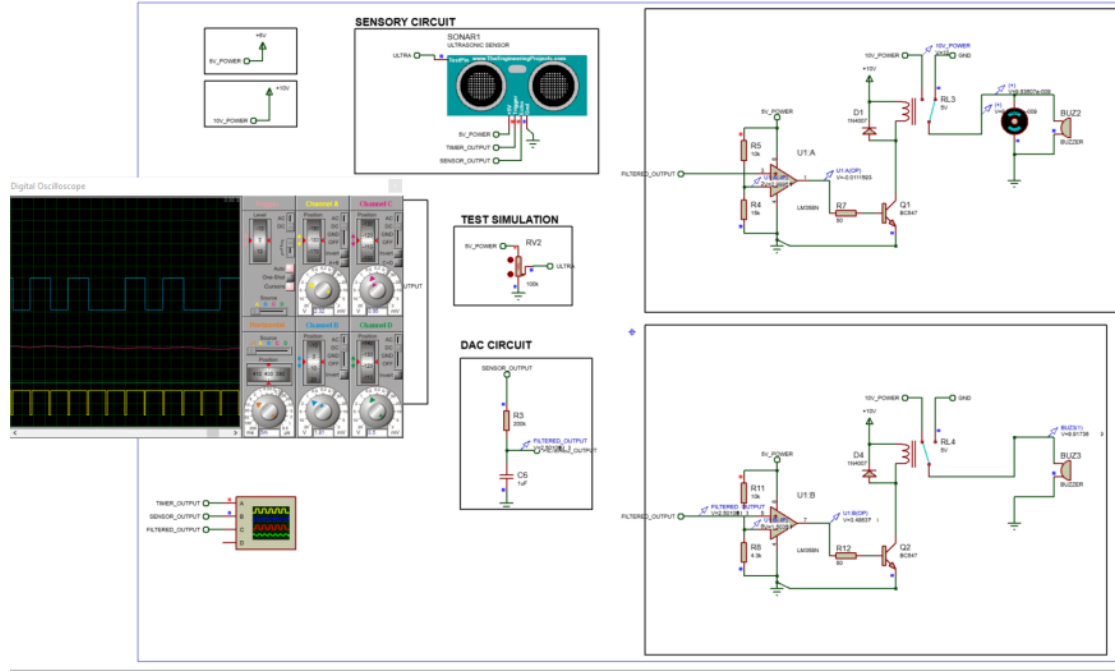


Figure 18: Diagram showing the circuit response when an object is detected at 1-metre range.

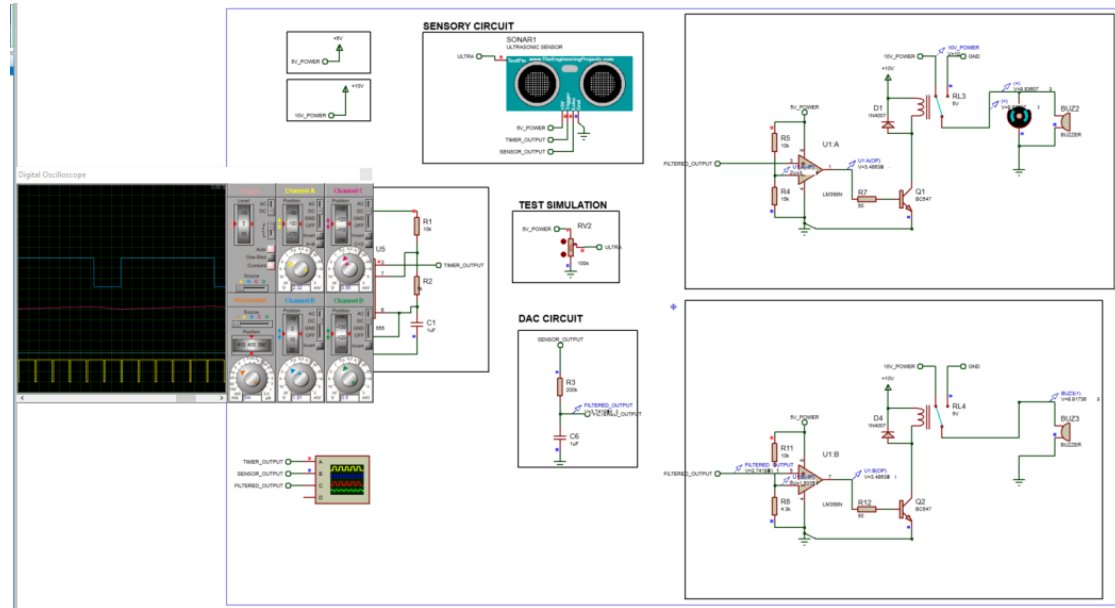


Figure 19: Diagram showing the circuit response when an object is detected at 3-metre range.

Conclusion

This research paper presents the design, simulation, and analysis of an aiding system for visually impaired individuals using a handheld, rechargeable stick which operates on proximity sensors. The main objective was to detect objects at two distinct distances: 1 metre and 3 metres. The device is handheld and rechargeable, and intended for daily use. The system was designed to be easy to use.

This project addresses problems faced by visually impaired individuals such as requiring secondary personnel for mobility and navigation in unknown surroundings. Users will be able to navigate unknown spaces with a navigation system that detects obstacles well in advance. This design demonstrates significant potential for further improvement; challenges encountered during implementation can be overcome with the appropriate time and resources.

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